“AI BASED RESUMING SCREENING SYSTEM”

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA0236- C PROGRAMMING FOR SOFTWARE DEVICES

to the award of the degree of

BACHELOR OF ENGINEERING

IN

B.E-CSE

Submitted by

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DECLARATION

We, **A.LIKITHASRI** of the **B.E-CSE,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **'AI BASED RESUMING SCREENING SYSTEM'** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place:Chennai

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Signature of the Students with Names

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**AI BASED RESUME SCREENING SYSTEM** " has been carried out by **A.LIKITHASRI** under the supervision of **DR.B.Gunasundari** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Submitted for the Project work Viva-Voce held on _____

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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Signature With Student Name

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ABSTRACT

In the modern business environment, automation has become an essential part of improving efficiency, accuracy, and speed of operations. However, many small retail shops and local businesses still rely on traditional handwritten billing systems. These manual systems are time-consuming, prone to calculation errors, and inefficient for managing sales and inventory records. One of the major problems faced by shop owners is the manual calculation of GST, which often results in mistakes and customer dissatisfaction.

The project titled **"AI BASED RESUME SCREENING SYSTEM"** is developed to overcome all these limitations. This system automates the entire billing process by allowing shopkeepers to add items to a bill, calculate GST automatically, store customer and sales data digitally, manage product inventory, and generate professional PDF invoices for customers. The entire system is developed using **C PROGRAMMING**, which serves as the core technology for all functionalities.

This system helps small and medium-scale businesses such as grocery stores, medical shops, supermarkets, and stationery stores to improve their billing process. It reduces manual work, saves time, improves accuracy, and gives a professional look to the business. The automated billing system also helps in maintaining proper sales history and inventory records for future analysis and decision-making.

CHAPTER 1: INTRODUCTION

Background Information

In today's competitive job market, organizations receive a large number of job applications for a single position. Manually reviewing each resume is time-consuming, inefficient, and often leads to human errors or bias. Recruiters may overlook qualified candidates due to the high volume of applications or lack of proper filtering mechanisms. At the same time, job seekers face challenges in ensuring their resumes meet the expectations of recruiters and automated systems.

With the advancement of Artificial Intelligence (AI), resume screening has become more efficient and accurate. AI-based systems can analyze resumes, identify relevant skills, and match candidates with job requirements. These systems help automate the initial screening process, reducing workload and improving the quality of candidate selection. This project focuses on developing an AI-based resume screening system that simplifies and enhances the recruitment process.

Project Objectives

The primary objective of this project is to develop an AI-based resume screening system that can automatically evaluate resumes based on predefined criteria. The system aims to reduce manual effort by filtering and shortlisting candidates efficiently.

Another objective is to improve the accuracy and consistency of the screening process by analyzing candidate details such as skills, education, and experience. The system also aims to provide a fair and unbiased evaluation by applying the same criteria to all applicants.

Additionally, the project demonstrates the implementation of programming concepts in C, including structures, file handling, and functions, while incorporating basic AI logic for decision-making.

Significance

This project is significant because it addresses a major challenge in recruitment: handling a large number of resumes efficiently. By automating the screening process, the system saves time and effort for recruiters while ensuring that qualified candidates are not overlooked.

From a technological perspective, the project highlights the practical application of AI in real-world scenarios. It shows how intelligent systems can improve decision-making and increase efficiency.

From a societal perspective, the system promotes fairness and transparency in hiring by reducing human bias and ensuring that all resumes are evaluated based on the same criteria.

Scope

The scope of this project includes the development of a system that can analyze and screen resumes based on specific parameters such as skills, qualifications, and keywords. The system collects candidate data, processes it, and generates results indicating whether a candidate is suitable or not.

The project focuses on basic AI logic and structured programming using the C language. It does not include advanced features such as machine learning models, real-time job portal integration, or large-scale database management.

The system is intended for educational purposes and demonstrates how automation can be applied in the recruitment process.

Methodology Overview

The methodology followed in this project involves collecting resume data from users and processing it using structured programming techniques. The system stores candidate information using data structures and compares it with predefined job requirements.

Based on matching criteria such as skills and qualifications, the system evaluates each resume and determines its suitability. Logical conditions and simple AI-based decision rules are applied to filter candidates.

The final output displays the screening result, which helps in identifying suitable candidates efficiently. The development process focuses on simplicity, accuracy, and effective implementation of core programming concepts.

CHAPTER 2: PROBLEM STATEMENT, OBJECTIVES AND SCOPE

Description of the Problem

In the modern recruitment process, organizations often receive hundreds or even thousands of resumes for a single job opening. Manually reviewing and shortlisting candidates from such a large pool is a time-consuming and inefficient task. Recruiters may not be able to carefully analyze every resume due to time constraints, leading to the possibility of missing qualified candidates. Additionally, manual screening can be inconsistent, as different recruiters may apply different criteria or judgments while evaluating resumes. Another major issue is the lack of standardization in resume formats. Candidates present their information in various styles, making it difficult for recruiters to quickly extract relevant details such as skills, education, and experience. This increases the complexity of the screening process and reduces overall efficiency.

Evidence of the Problem

Studies and industry observations show that recruiters spend only a few seconds reviewing each resume during the initial screening stage. Due to the high volume of applications, many resumes are rejected without detailed evaluation. This indicates the need for a faster and more reliable screening system. In addition, many organizations are increasingly adopting automated tools to handle recruitment processes. The growing demand for such systems highlights the limitations of traditional manual screening methods. The inefficiency, time consumption, and possibility of human error clearly demonstrate the existence of the problem.

Stakeholders

The primary stakeholders affected by this problem include recruiters, job seekers, and organizations. Recruiters are directly impacted as they are responsible for screening large numbers of resumes within limited time. Job seekers are also affected because their chances of being selected depend on how effectively their resumes are reviewed. Organizations rely on efficient recruitment processes to hire qualified candidates and maintain productivity. Human resource departments also play a key role, as their performance is directly influenced by how effectively the screening process is carried out.

Supporting Data/Research

Research in recruitment practices indicates that automated screening systems significantly reduce the time required for initial candidate evaluation. Many companies use Applicant Tracking Systems (ATS) to filter resumes based on keywords and predefined criteria. These systems help improve accuracy and consistency in candidate selection. Furthermore, studies show that structured and automated processes reduce bias and improve fairness in hiring. By using technology to analyze resumes, organizations can ensure that candidates are evaluated based on relevant qualifications and skills rather than subjective judgment.

Chapter 3: Solution Design and Implementation

Development and Design Process

The development of the AI-based resume screening system follows a structured approach to ensure efficiency and accuracy. Initially, the system requirements are identified, including the type of input data such as candidate details, skills, education, and experience. Based on these requirements, the system is designed using structured programming principles. The design phase includes defining data structures, creating input methods, and planning the logic for screening resumes. The implementation phase involves writing modular code using functions, where each function performs a specific task such as data collection, processing, and evaluation. Proper testing and debugging are carried out to ensure that the system functions correctly and produces accurate screening results.

Tools and Technologies Used

The system is developed using the C programming language, which provides a strong foundation for implementing structured programming concepts. Key concepts used in this project include structures for storing candidate data, file handling for saving and retrieving information, and functions for modular programming. The program is executed using compilers such as GCC or Turbo C. These tools help in compiling, running, and testing the program efficiently. The use of C language ensures simplicity, speed, and better control over system resources.

Solution Overview

The AI-based resume screening system is designed to automate the process of evaluating resumes. The system collects input data from users, including personal details, educational qualifications, skills, and experience. This data is stored and processed using appropriate data structures. The system then compares the candidate's information with predefined job requirements using logical conditions. Based on this comparison, the system determines whether the candidate meets the required criteria. The final output displays the screening result, indicating whether the candidate is selected or rejected. This automated process reduces manual effort and ensures consistency in evaluation.

Engineering Standards Applied

During the development of the project, basic software engineering principles and standards are followed to ensure quality and reliability. Structured programming is used to maintain clarity and organization in the code. Modular design is applied by dividing the program into smaller functions, making it easier to understand and maintain. Proper naming conventions and indentation are followed to improve readability. Input validation techniques are also implemented to ensure that the system handles data correctly and avoids errors during execution.

Solution Justification

The use of structured programming and basic AI logic makes the system efficient and reliable for resume screening. By applying standard programming practices, the project ensures that the code is easy to debug, maintain, and extend in the future. The automated screening process improves accuracy by applying consistent criteria to all candidates, reducing the chances of human error and bias. Overall, the inclusion of these standards enhances the performance and effectiveness of the system, making it a practical solution for handling resume screening tasks.

Chapter 4: Results and Recommendations

Evaluation of Results

The AI-based resume screening system successfully performs the task of evaluating candidate resumes based on predefined criteria such as skills, qualifications, and experience. The system is able to process user input efficiently and generate accurate screening results indicating whether a candidate is suitable for a given role. The automated approach significantly reduces the time required for initial screening compared to manual methods. It also ensures consistency, as all resumes are evaluated using the same conditions and logic. Overall, the system demonstrates reliable performance and achieves its objective of simplifying the resume screening process.

Challenges Encountered

During the implementation of the project, several challenges were encountered. One of the main difficulties was designing the logic for comparing candidate data with job requirements in an effective way. Handling user input correctly and avoiding errors during data entry was also a challenge. Additionally, formatting and organizing the output in a clear and structured manner required careful planning. File handling operations sometimes caused issues related to data storage and retrieval. These challenges were overcome by debugging the code, using proper validation techniques, and applying structured programming methods to improve accuracy and reliability.

Possible Improvements

Although the system performs its basic functions effectively, there are several areas where improvements can be made. The system can be enhanced by adding a graphical user interface (GUI) to make it more user-friendly. More advanced AI techniques, such as machine learning and natural language processing, can be integrated to analyze resumes more intelligently. The system can also be expanded to support multiple job roles and dynamic criteria instead of fixed conditions. Additionally, integrating the system with online job portals or databases can improve its practical usability in real-world scenarios.

Recommendations

For future development, it is recommended to enhance the system by incorporating advanced technologies that improve accuracy and scalability. Implementing machine learning algorithms can help the system learn from data and make better screening decisions over time. It is also recommended to develop a web-based or mobile-based version of the system to increase accessibility. Further research can be conducted to improve keyword matching techniques and reduce bias in automated screening. These improvements will make the system more powerful, efficient, and suitable for real-world recruitment processes.

Chapter 5: Reflection on Learning and Personal Development

Key Learning Outcomes

The development of the AI-based resume screening system provided a strong understanding of how programming concepts can be applied to solve real-world problems. Through this project, knowledge of C programming was significantly improved, especially in areas such as structures, functions, string handling, and file operations. The use of a structure to store candidate details helped in understanding how data can be organized efficiently. The implementation of the calculate_score function demonstrated how logical conditions and decision-making can be applied to simulate basic Artificial Intelligence behavior. The concept of keyword matching using string functions like strstr() helped in understanding how systems can analyze textual data. Additionally, the project enhanced problem-solving and critical thinking skills by requiring the design of a scoring system based on multiple conditions such as skills and experience.

Challenges Encountered and Overcome

During the development of the project, several challenges were faced. One of the main difficulties was implementing the logic for skill-based evaluation, as it required careful handling of string inputs and matching keywords accurately. There were also challenges in handling user input correctly, especially when reading full strings with spaces, which required modifications in input methods. Managing file operations such as opening, writing, and closing files was another challenge, as errors could occur if not handled properly. Debugging the program to ensure correct score calculation and output formatting required patience and attention to detail. These challenges helped in improving coding skills, logical thinking, and the ability to troubleshoot errors effectively. Overcoming these issues increased confidence in handling programming tasks independently.

Application of Engineering Standards

The project followed fundamental programming standards to ensure clarity, efficiency, and maintainability. Structured programming was applied by dividing the program into logical sections such as data representation, evaluation, and storage. The use of functions improved modularity and reduced code repetition. Proper indentation and naming conventions were followed to enhance readability. Input validation and careful handling of strings ensured that

the system operates reliably. File handling techniques were used efficiently to store results, reflecting real-world software practices. These standards contributed to the overall quality and organization of the project.

Insights into the Industry

This project provided an understanding of how resume screening is performed in real-world recruitment systems. It highlighted the importance of automation in handling large volumes of applications efficiently. The use of keyword matching in the project reflects how many companies use Applicant Tracking Systems (ATS) to filter resumes based on specific criteria. It also showed how even simple AI logic can significantly improve efficiency and accuracy in decision-making processes. This experience created interest in exploring advanced technologies such as machine learning and natural language processing, which are widely used in modern recruitment systems.

Conclusion of Personal Development

Overall, the project contributed greatly to personal and academic growth. It strengthened programming skills, improved problem-solving ability, and increased confidence in developing practical applications. The experience of building a complete system from input to output provided a deeper understanding of software development. It also emphasized the importance of continuous learning and adapting to new technologies. This project has helped in preparing for future academic work and professional opportunities in the field of computer science and technology.

Chapter 6: Conclusion

The AI-based resume screening system developed in this project effectively addresses the challenges associated with manual resume evaluation in the recruitment process. By automating the screening of candidate resumes using predefined criteria such as skills and experience, the system significantly reduces the time and effort required by recruiters. The use of keyword matching and a structured scoring mechanism ensures that each candidate is evaluated consistently, thereby minimizing human bias and improving the overall fairness of the selection process.

This project successfully demonstrates the practical implementation of core programming concepts in C, including structures for data organization, functions for modular design, string handling for text analysis, and file operations for data storage. The integration of these concepts into a single system highlights how theoretical knowledge can be applied to develop meaningful real-world applications. Additionally, the use of basic AI-like logic in decision-making provides an introduction to how intelligent systems operate in modern industries.

The system also emphasizes the importance of automation in handling large volumes of data efficiently. In real-world scenarios, companies receive numerous applications for job positions, making manual screening impractical. This project provides a simplified model of how such processes can be optimized using technology. Although the current system uses basic logic, it lays a strong foundation for future enhancements.

There is significant scope for further improvement, including the integration of advanced technologies such as machine learning algorithms and natural language processing techniques to analyze resumes more intelligently. Features such as dynamic job matching, ranking systems, and web-based interfaces can also be added to increase usability and scalability.

In conclusion, the project not only fulfills its objective of developing an AI-based resume screening system but also provides valuable learning experience in problem-solving, programming, and system design. It demonstrates the potential of combining programming with AI concepts to create efficient and impactful solutions, thereby preparing for future advancements in the field of technology.

C PROGRAMMING CODE FOR THE WEBSITE:

Code Blame 53 lines (42 loc) · 1.26 KB

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

// MODULE 1: DATA REPRESENTATION
struct Candidate {
    char name[50];
    char skills[100];
    int experience;
    int score;
};

// MODULE 2: EVALUATION LOGIC
int calculate_score(char skills[], int exp) {
    int score = 0;
    // Simple keyword matching (AI-lite logic)
    if (strstr(skills, "C")) score += 30;
    if (strstr(skills, "Python")) score += 30;
    if (strstr(skills, "Data Structures")) score += 20;

    // Experience threshold
    if (exp >= 3) score += 20;
    else if (exp >= 1) score += 10;

    return score;
}

int main() {
    struct Candidate c;
    FILE *fptr;

    // Input
    printf("Enter Name: ");
    scanf("%s", c.name);
    printf("Enter Skills: ");
    scanf(" %[^\n]s", c.skills);
    printf("Enter Years of Experience: ");
    scanf("%d", &c.experience);

    c.score = calculate_score(c.skills, c.experience);

    // MODULE 3: STORAGE
    fptr = fopen("EvaluationReport.txt", "a");
    if(fptr == NULL) return 1;

    fprintf(fptr, "Candidate: %s | Score: %d\n", c.name, c.score);
    fclose(fptr);

    printf("\nAnalysis Complete. Score: %d%%\n", c.score);
    printf("Report saved to EvaluationReport.txt\n");

    return 0;
}
```

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